

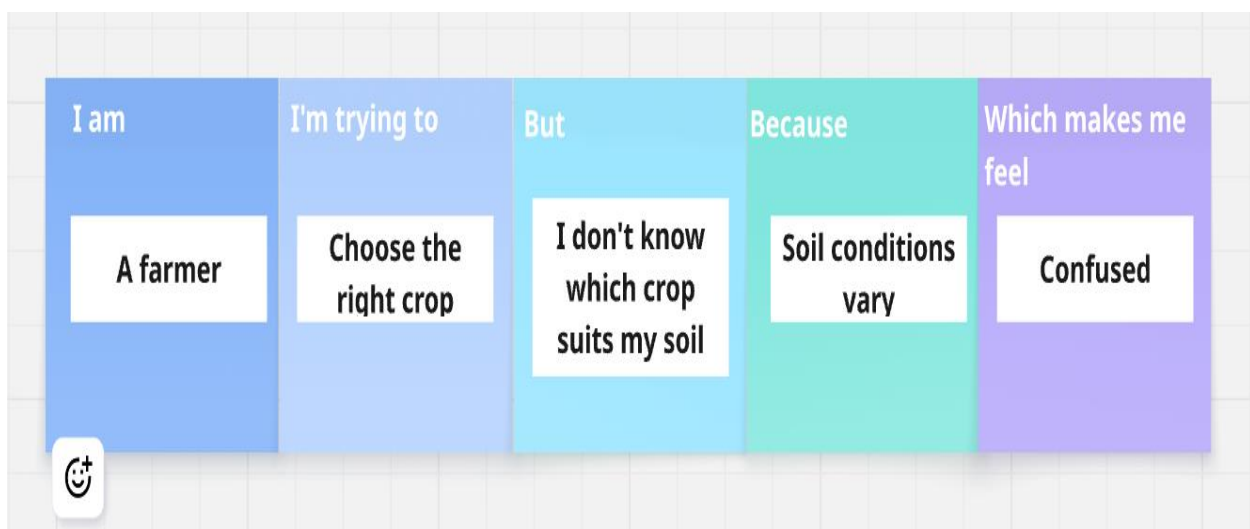
Ideation Phase

Define the Problem Statements

Date	26 June 2026
Project Name	OptiCrop
Maximum Marks	2 Marks

Customer Problem Statement Template:

The Customer Problem Statement for the **OptiCrop – Smart Agricultural Production Optimization Engine** project helps identify the key challenges faced by farmers in making informed agricultural decisions. It focuses on understanding farmers' needs, difficulties, and expectations regarding crop selection, soil nutrient management, and resource utilization. By clearly defining these problems, the project aims to develop an intelligent solution that provides accurate crop recommendations based on soil nutrients, temperature, humidity, pH, rainfall, and other environmental factors. Understanding the customer's perspective enables the team to design a solution that improves crop yield, reduces resource wastage, and supports sustainable farming practices.



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Choose the most suitable crop based on my soil and environmental conditions.	I am unable to identify the best crop for my land.	I don't have access to accurate analysis of soil nutrients, pH, rainfall, temperature, and humidity.	Uncertain about my farming decisions and worried about reduced crop yield.
PS-2	A farmer	Improve crop productivity while minimizing the use of fertilizers and other resources.	I often apply fertilizers based on guesswork instead of scientific recommendations .	I lack real-time insights into soil nutrient levels and environmental conditions.	Frustrated because my costs increase while crop yield and profits decrease.